

Q1) SVM

10

$$6x_1 + 8x_2 - 48 = 0$$

$$P = (10, 2)$$

$$Q = (3, 3)$$

a) $6 \cdot 10 + 8 \cdot 2 - 48 = 28$

— P is above the boundary

~~28~~ $6 \cdot 3 + 8 \cdot 3 - 48 = -6$

— Q is below the boundary

$$\|W\| = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = 10$$

b) P distance = $\frac{28}{10} = 2.8$

$$\text{Distance}(x) = \frac{|W \cdot x + b|}{\|W\|}$$

Q distance = $\frac{6}{10} = 0.6$

c) more likely for Q to be a support boundary, because it's closer to the boundary.